

Economics S5040: Dynamic Stochastic Modeling

Problem Set 5 — The McCall Job Search Model

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Learning Objectives

By the end of this problem set, you should be able to:

- Recognize an optimal-stopping problem.
- Write a Bellman equation when the choice is “stop or continue.”
- Understand that dynamic programming does not require an explicit discount factor.
- Derive and interpret a cutoff policy.
- Solve the McCall job-search model using value-function iteration.
- Interpret the reservation wage.
- Analyze how the value of waiting affects optimal stopping.

Problem 1. The Parking Problem: A Simple Optimal-Stopping Model

You are driving down a one-way street toward your office. Parking spaces occur at regular intervals.

Your office is next to the best possible parking space. Parking there gives you a loss of 0. Parking one space away gives a loss of 1, parking two spaces away gives a loss of 2, and so forth.

You can see only the parking space you are currently passing. Each parking space is independently open with probability p , where

$$0 < p < 1.$$

When you encounter an open parking space, you must decide immediately whether to park or continue. Once you pass a space, you cannot return to it.

If you pass your office without finding a parking space, you continue down the street and park in the first available space.

(a) Why is this an optimal-stopping problem?

What are the two possible actions when you encounter an open space? What does “stop” mean? What does “continue” mean? Explain why rejecting an open parking space is an irreversible decision.

(b) A useful infinite sum

Before solving the parking problem, consider the infinite series

$$S = p + 2p^2 + 3p^3 + 4p^4 + \cdots .$$

Show that

$$S = \frac{p}{(1-p)^2}.$$

Hint: Multiply S by p :

$$pS = p^2 + 2p^3 + 3p^4 + \cdots .$$

Now subtract pS from S . What familiar geometric series remains?

(c) Expected loss after passing the office

Suppose you reach the office and its parking space is occupied. You now know that you will take the first open parking space you encounter.

Let $q = 1 - p$. Show that the probability that the first available space is one space beyond the office is p , two spaces beyond the office is qp , three spaces beyond the office is q^2p , and so forth.

Therefore the expected loss, conditional on the office space being occupied, is

$$p + 2qp + 3q^2p + 4q^3p + \cdots .$$

Use the result from part (b), with q replacing p , to show that this expected loss is

$$\frac{1}{p}.$$

Explain the intuition. What happens to the expected walking distance as p approaches zero? What happens as p approaches one?

(d) The value at the office

Let $V(0)$ be expected loss immediately before observing whether the parking space at the office is available.

If it is available, you park and incur zero loss. If it is occupied, you pass the office and incur the expected loss calculated in part (c).

Show that

$$V(0) = \frac{1-p}{p}.$$

(e) The Bellman equation

Now suppose you are n spaces before the office. Let $V(n)$ denote your expected loss before observing whether the current parking space is open.

If the space is occupied, you must continue. If the space is open, you have a choice: park now and incur loss n , or continue and receive expected loss $V(n - 1)$.

Show that the Bellman equation is

$$V(n) = p \min\{n, V(n - 1)\} + (1 - p)V(n - 1).$$

Why is there no discount factor in this Bellman equation?

(f) The stopping rule

Suppose the current space is open. Show that you should park if

$$n \leq V(n - 1),$$

and continue if

$$n > V(n - 1).$$

Explain why you should expect the optimal policy to have a cutoff form. In words, the policy should look like: “Ignore all available spaces until I get sufficiently close to the office. After that point, take the first available space.”

(g) Find the cutoff

Show that the stopping condition can ultimately be written as

$$(1 - p)^n \geq \frac{1}{2}.$$

Therefore the cutoff is determined by

$$n \leq \frac{\log(1/2)}{\log(1 - p)}.$$

Because n must be an integer, explain carefully how you would convert this expression into a parking rule.

Calculate the optimal rule for

$$p = 0.10, \quad p = 0.20, \quad p = 0.50, \quad p = 0.80.$$

Does the relationship between p and the cutoff make economic sense?

(h) Dynamic programming without discounting

There is no β in this problem. Explain why this is nevertheless a dynamic programming problem. In particular, distinguish between:

1. discounting future outcomes, and
2. making a current decision that changes the state and therefore changes the set of future opportunities.

What role does the probability $1 - p$ play in the recursion? Why should it not be interpreted as a discount factor?

Problem 2. The McCall Job Search Model

Now consider an unemployed worker who receives one wage offer each period. The possible wage offers are

$$w_i = 10, 10.5, 11, \dots, 60.$$

Suppose each wage is equally likely. The worker receives unemployment compensation

$$c = 12$$

and discounts future utility using

$$\beta = 0.95.$$

An accepted job lasts forever.

(a) Stop or continue

Explain why the McCall model is also an optimal-stopping problem. What corresponds to “parking” in the parking problem? What corresponds to “continuing past the parking space”?

If the worker accepts wage w , show that the value is

$$A(w) = \frac{w}{1 - \beta}.$$

If the worker rejects the wage, the value is

$$C = c + \beta E[V(W')].$$

Explain why C does not depend on the current wage offer.

(b) Bellman equation

Show that

$$V(w) = \max\{A(w), C\}.$$

Explain economically what each side of the maximum represents.

(c) Reservation wage

Show that there must be a reservation wage $\bar{w}$ such that

$$\text{accept if } w \geq \bar{w}, \quad \text{reject if } w < \bar{w}.$$

Using the equality between accepting and rejecting at the cutoff, show that

$$\bar{w} = (1 - \beta)C.$$

Compare this rule with the cutoff rule in the parking problem.

Problem 3. Solve the McCall Model Numerically

Construct the wage grid

$$w = 10, 10.5, \dots, 60.$$

Assume every wage has equal probability. Start with an initial guess for $V(w)$ and use value-function iteration to solve

$$V_{\text{new}}(w) = \max \left\{ \frac{w}{1 - \beta}, c + \beta E[V(W)] \right\}.$$

Continue iterating until

$$\max_w |V_{\text{new}}(w) - V(w)| < 10^{-10}.$$

- (a) Report the number of iterations required for convergence.
- (b) Calculate the continuation value C .
- (c) Calculate the reservation wage.
- (d) Plot $V(w)$ against w . On the same graph, plot the value of accepting and the value of rejecting. Indicate the reservation wage. Explain the kink in the value function.

Problem 4. The Value of Waiting

Return to the McCall model.

- (a) Increase unemployment compensation c . What happens to the continuation value? What happens to the reservation wage? Explain economically.
- (b) Increase β . What happens to the reservation wage? Why does greater patience make the worker more selective?
- (c) Change the wage-offer distribution so that high wages become more likely. What happens to the value of rejecting today's offer? What should happen to the reservation wage?
- (d) Relate these comparative statics to the parking problem. In the parking problem, what happens when p increases? In the job-search problem, what happens when the probability of receiving an attractive future wage increases? What common economic principle connects the two results?

Problem 5. Solve McCall Using the Continuation Value

Notice that the McCall problem has a special structure. The continuation value C is a scalar. Since

$$V(w) = \max \left\{ \frac{w}{1-\beta}, C \right\},$$

we can substitute this expression into the definition of C .

- (a) Show that C satisfies the scalar fixed-point equation

$$C = c + \beta E \left[\max \left\{ \frac{W}{1-\beta}, C \right\} \right].$$

- (b) Write Python code that iterates directly on C rather than on the entire value function V .
- (c) Verify that this produces the same continuation value and reservation wage as value-function iteration.
- (d) Which computational method is faster? Why can we reduce this particular dynamic program to a one-dimensional fixed-point problem?

Problem 6. Job Separation

Now suppose that an accepted job does not necessarily last forever. Each period, an employed worker loses the job with probability δ .

- (a) Write the value of accepting a job paying wage w . Remember that next period the worker may remain employed or become unemployed again.
- (b) Write the new Bellman equation.
- (c) Does the optimal policy still have a reservation-wage form? Explain.
- (d) How should an increase in δ affect the value of accepting a job? What should happen to the reservation wage? Give the economic intuition before doing any numerical calculation.

Final Question: What Did the Parking Problem Teach Us?

In a short paragraph, compare the parking problem with the McCall job-search model.

Both problems have a decision of the form

STOP versus CONTINUE.

But there is an important difference. McCall contains an explicit discount factor β . The parking problem does not.

Explain why both can nevertheless be solved using exactly the same dynamic-programming logic. In particular, discuss the following statement:

“Dynamic programming is not fundamentally about discounting. It is about making decisions today when today’s choice affects the opportunities available tomorrow.”